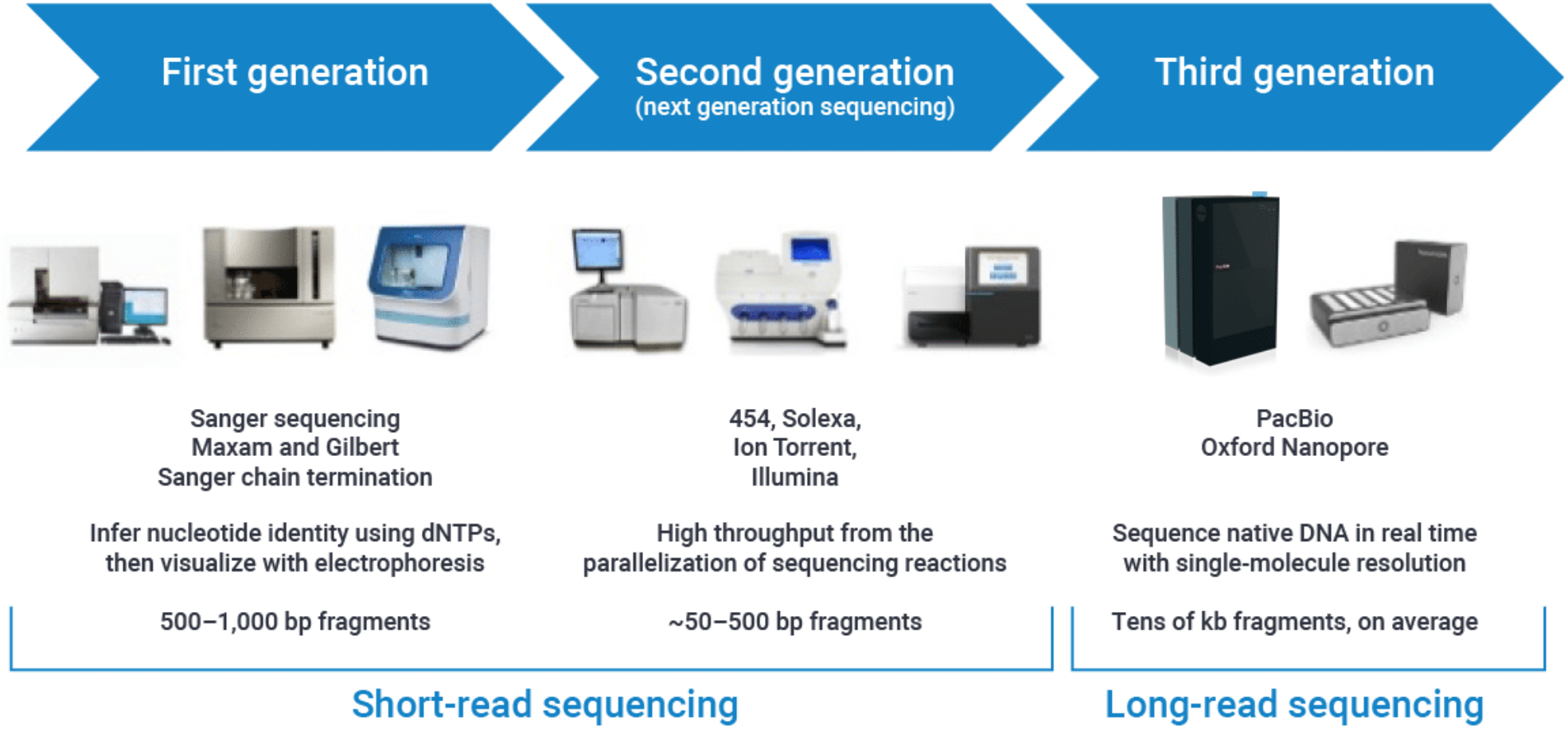
Sanger sequencing and phylogenetics

MMB-127

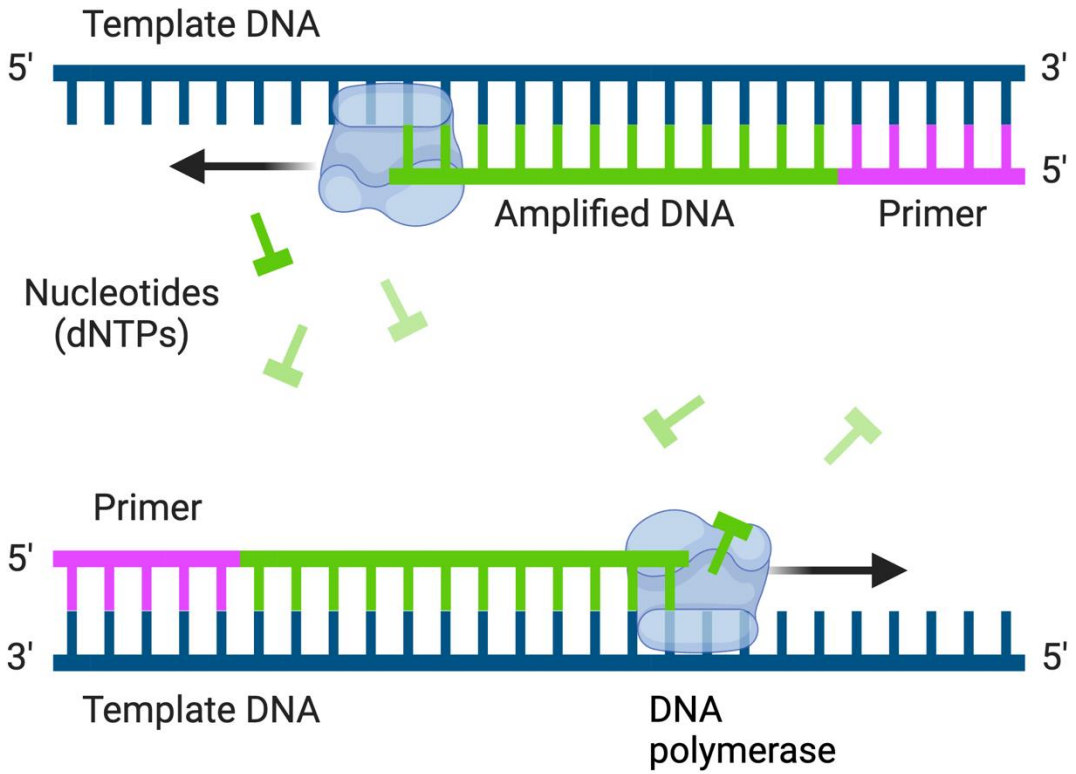
Evolution of sequencing



https://www.pacb.com/blog/the-evolution-of-dna-sequencing-tools/

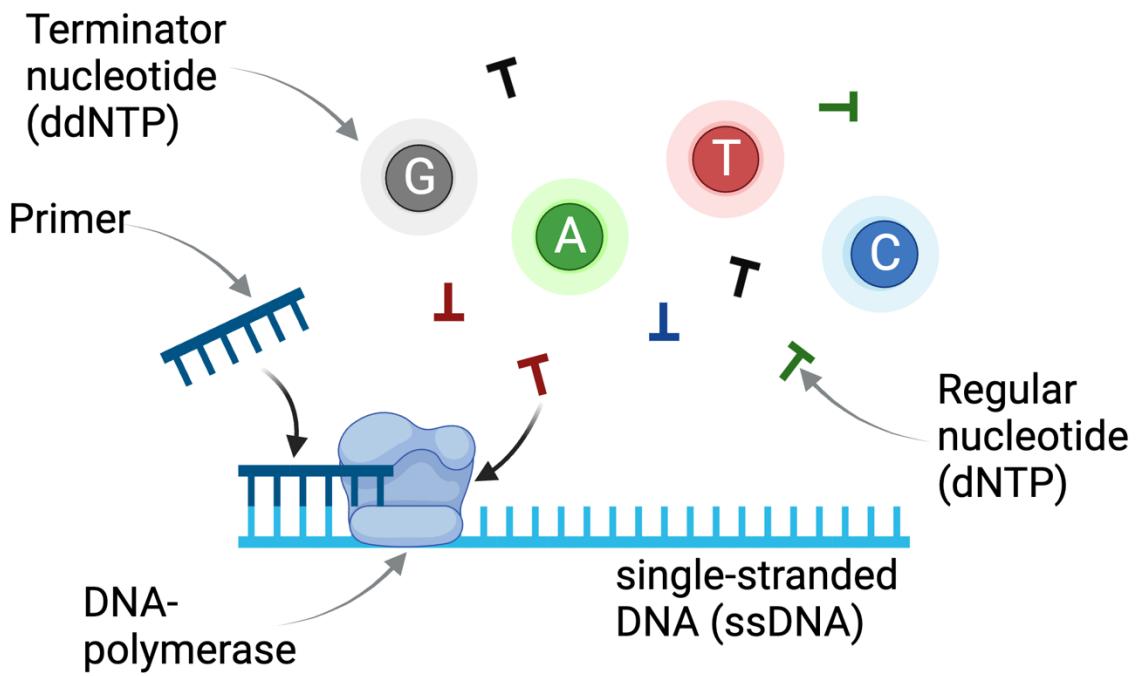
Sanger-sequencing

PCR - recap

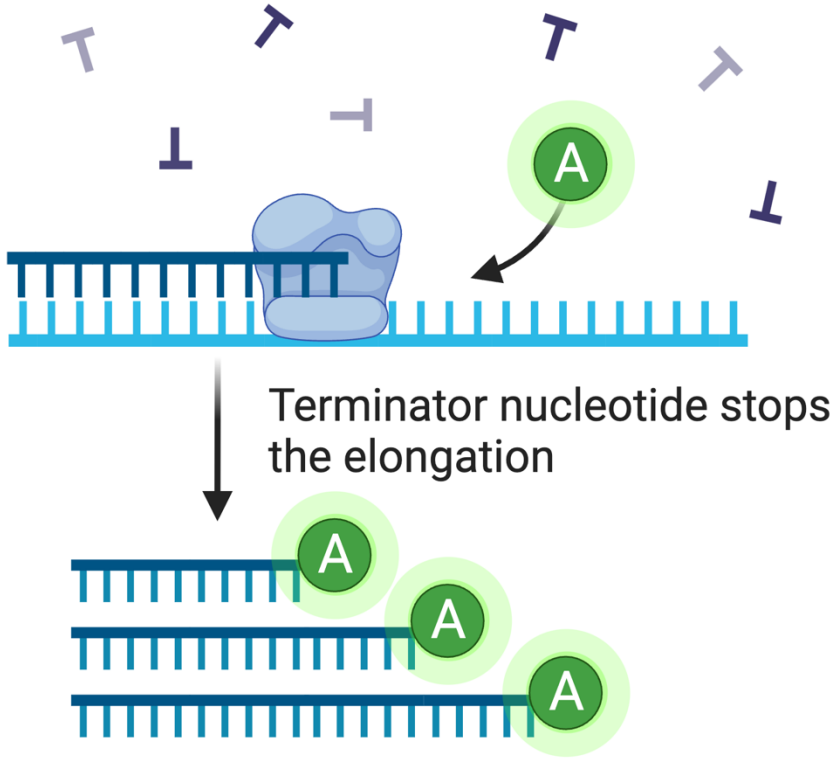

Template DNA
5'
3'
5'
Amplified DNA
Primer
Nucleotides
(dNTPs)
Primer
5'
3'
5'
Template DNA
DNA
polymerase

Terminator nucleotides

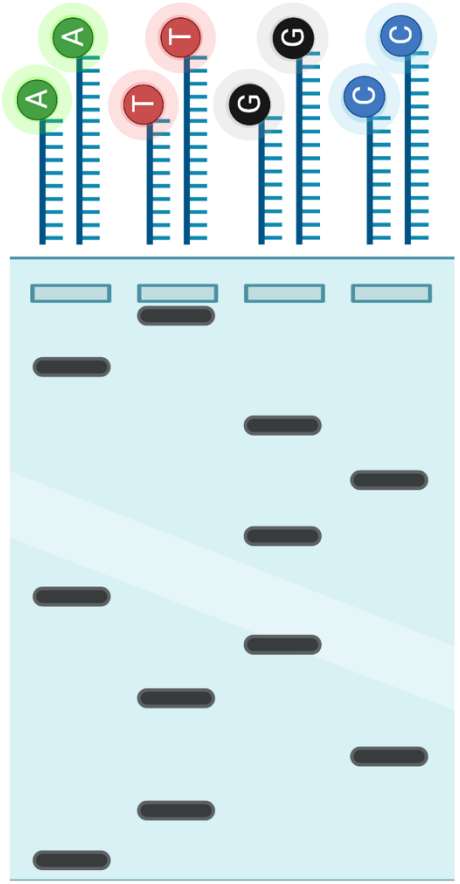
- Chain termination – dideoxy nucleotides (ddNTPs)



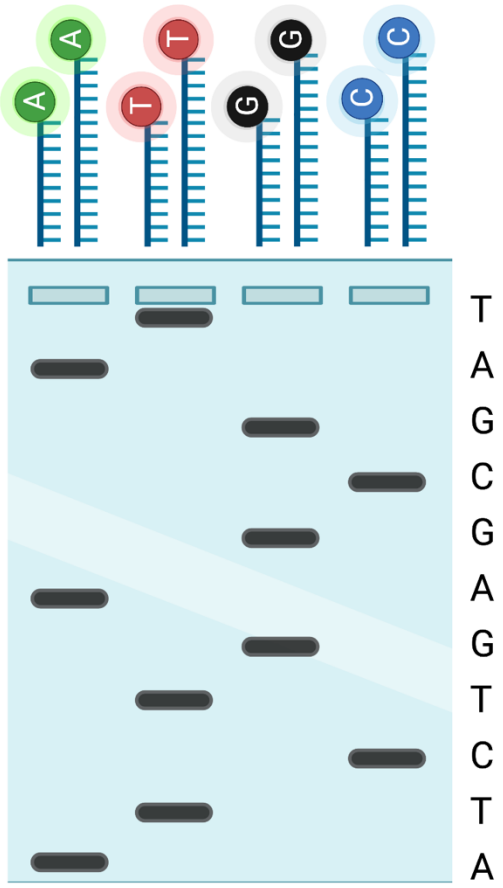
Terminator nucleotides



Sequencing gels



Sequencing gels

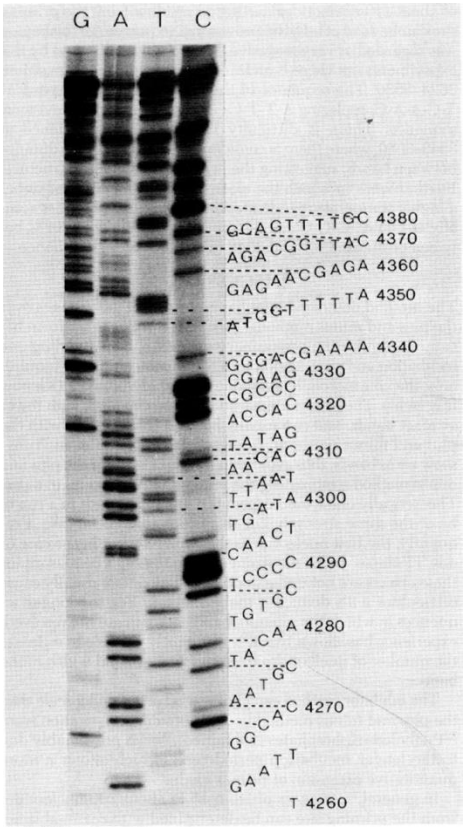


Proc. Natl. Acad. Sci. USA
Vol. 74, No. 12, pp. 5463–5467, December 1977
Biochemistry

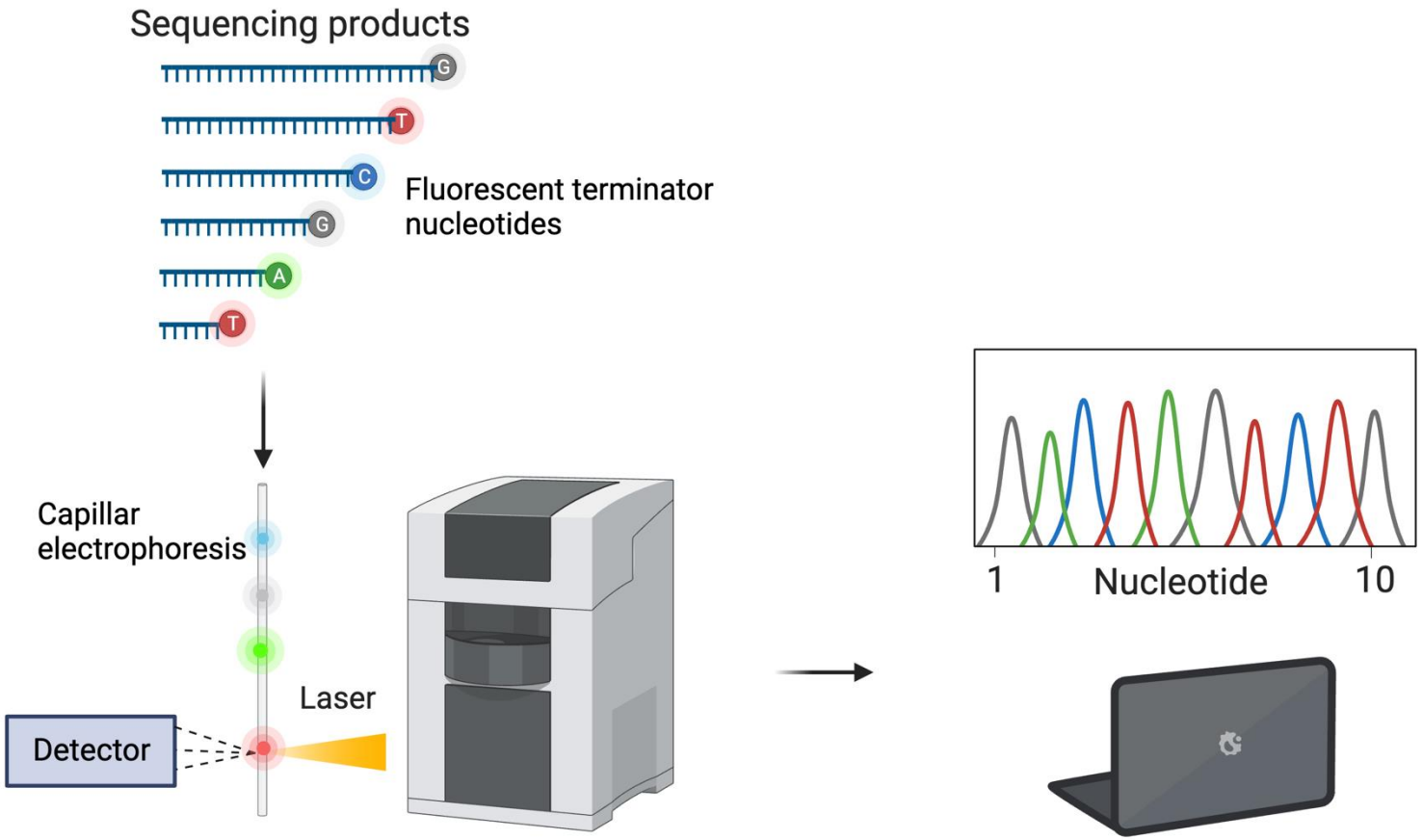
DNA sequencing with chain-terminating inhibitors

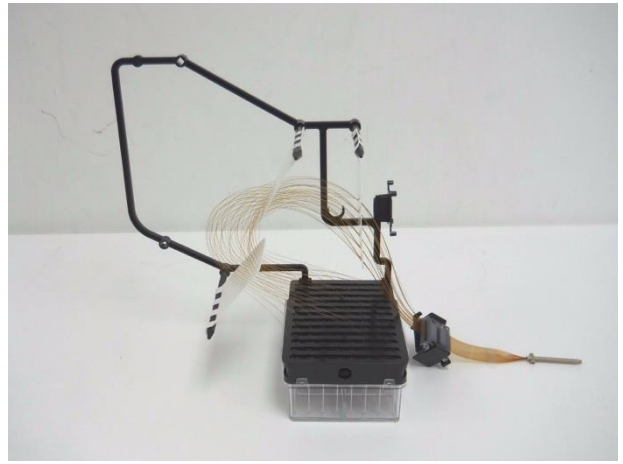
(DNA polymerase/nucleotide sequences/bacteriophage ϕX174)

F. SANGER, S. NICKLEN, AND A. R. COULSON



Automated Sanger-sequencing

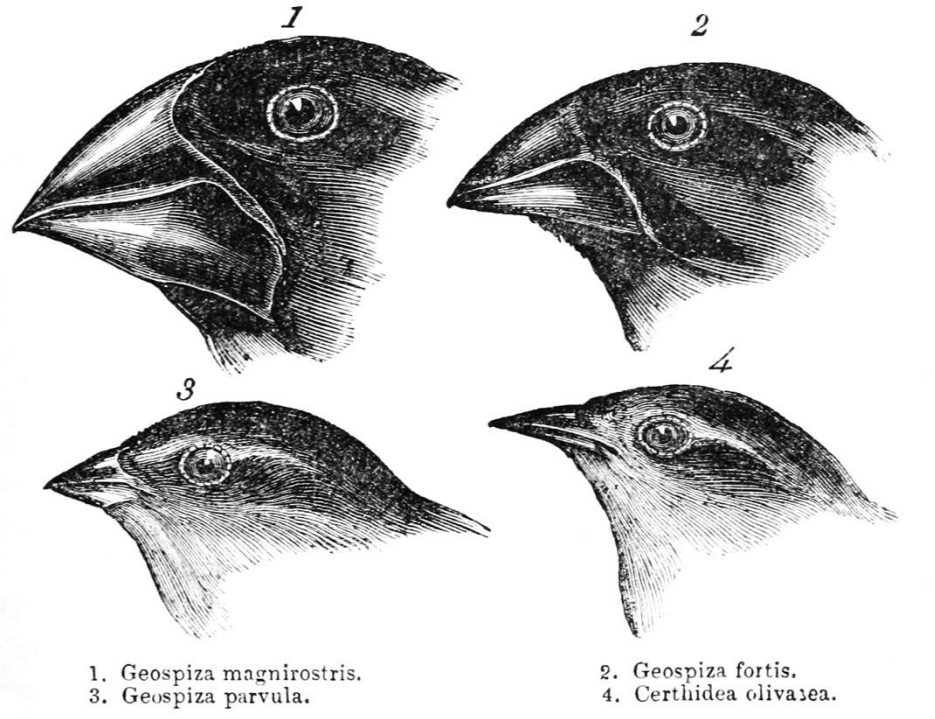




Phylogenetics

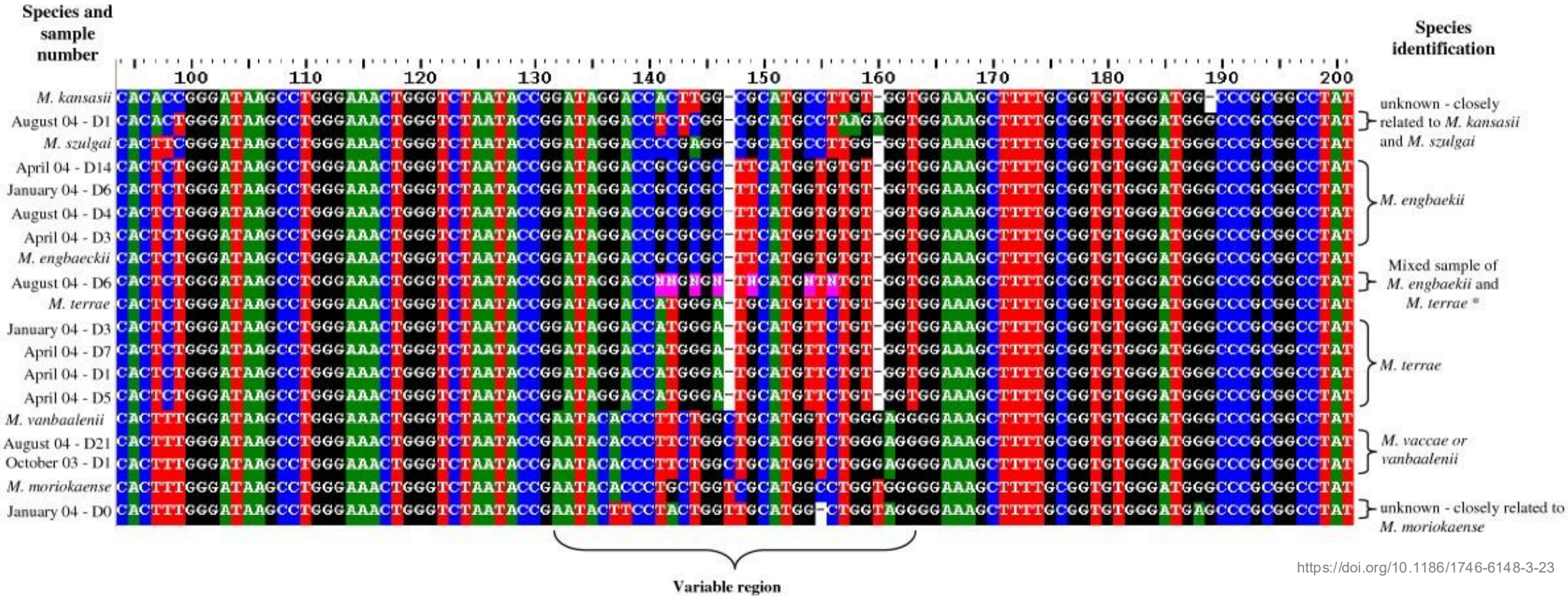
Phylogeny

- Evolutionary relationship of group of organisms
- Based on heritable traits
  - DNA sequence
  - Amino acid sequence
  - Morphology



https://en.wikipedia.org/wiki/Phylogenetics#/media/File:Darwin's_finches_by_Gould.jpg

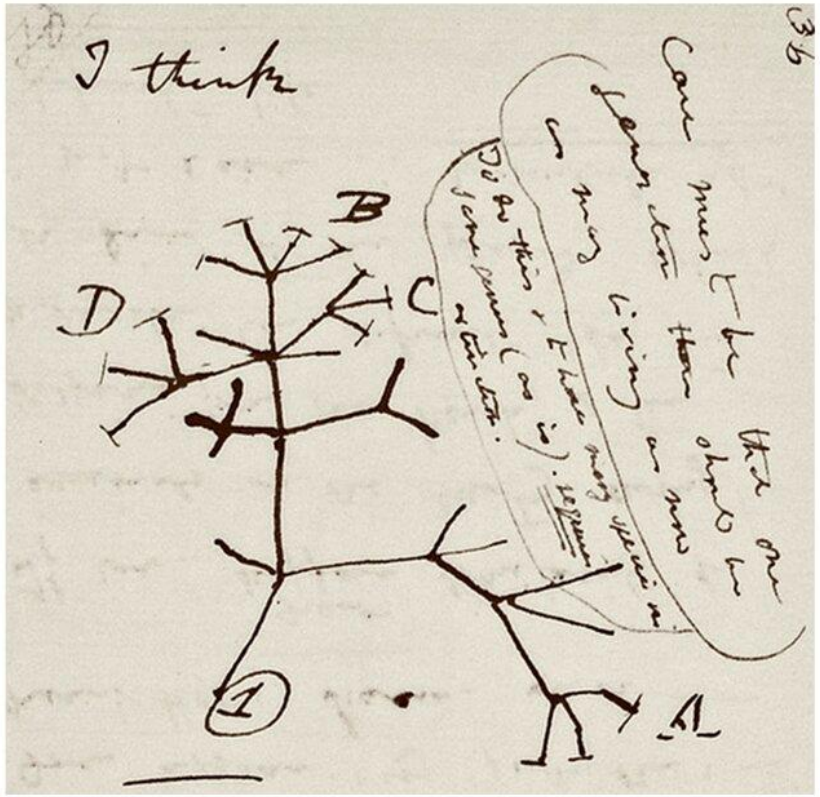
Multiple sequence alignment

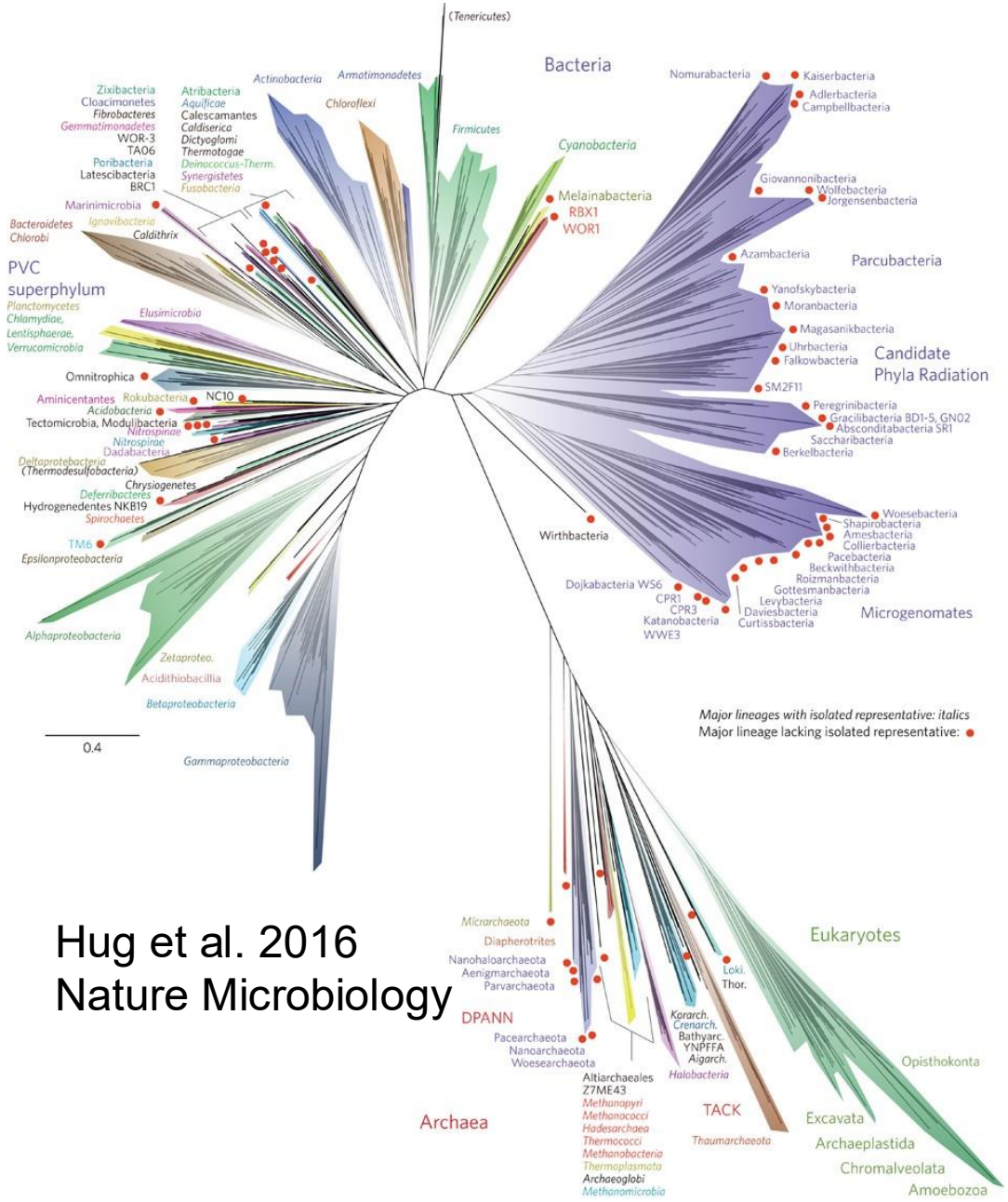


https://doi.org/10.1186/1746-6148-3-23

Phylogenetic trees

Darwin's notebook





Hug et al. 2016
Nature Microbiology

Phylogenetic trees

Scale

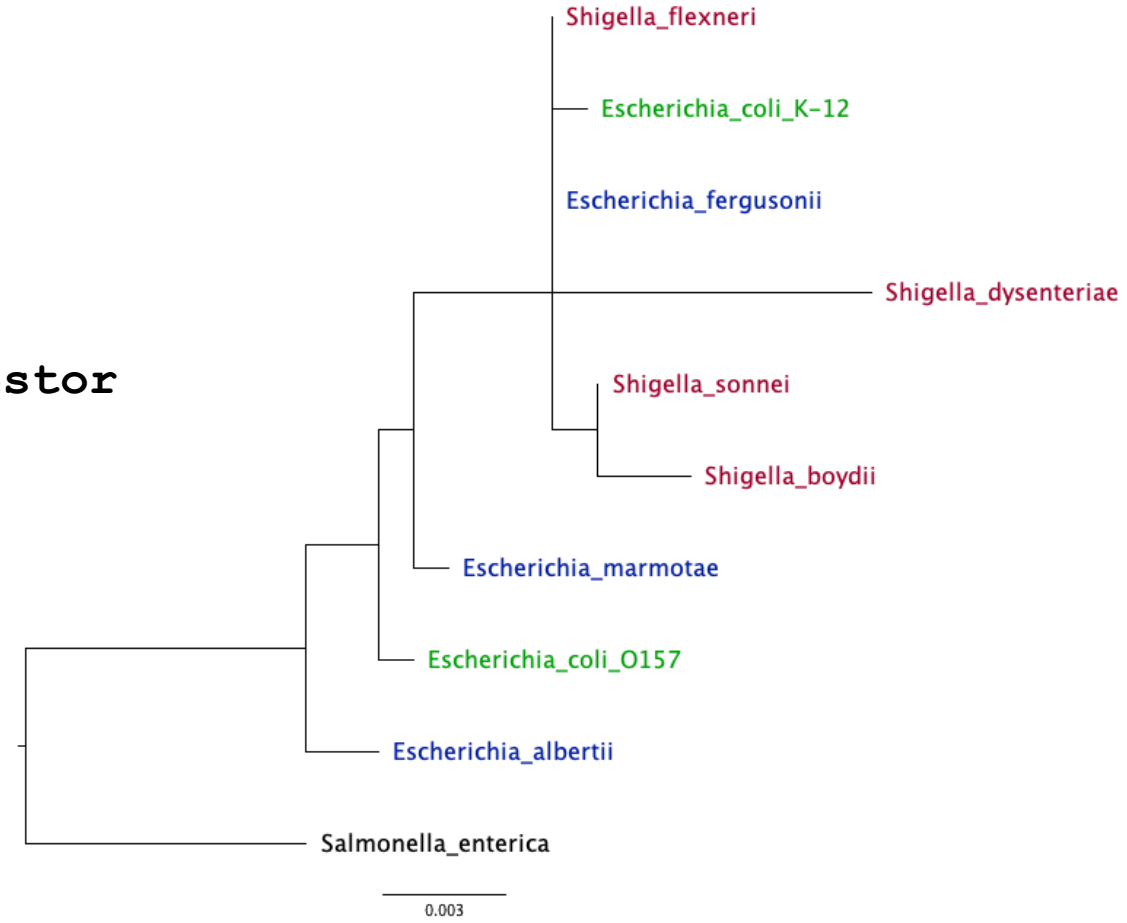
Node

Recent common ancestor

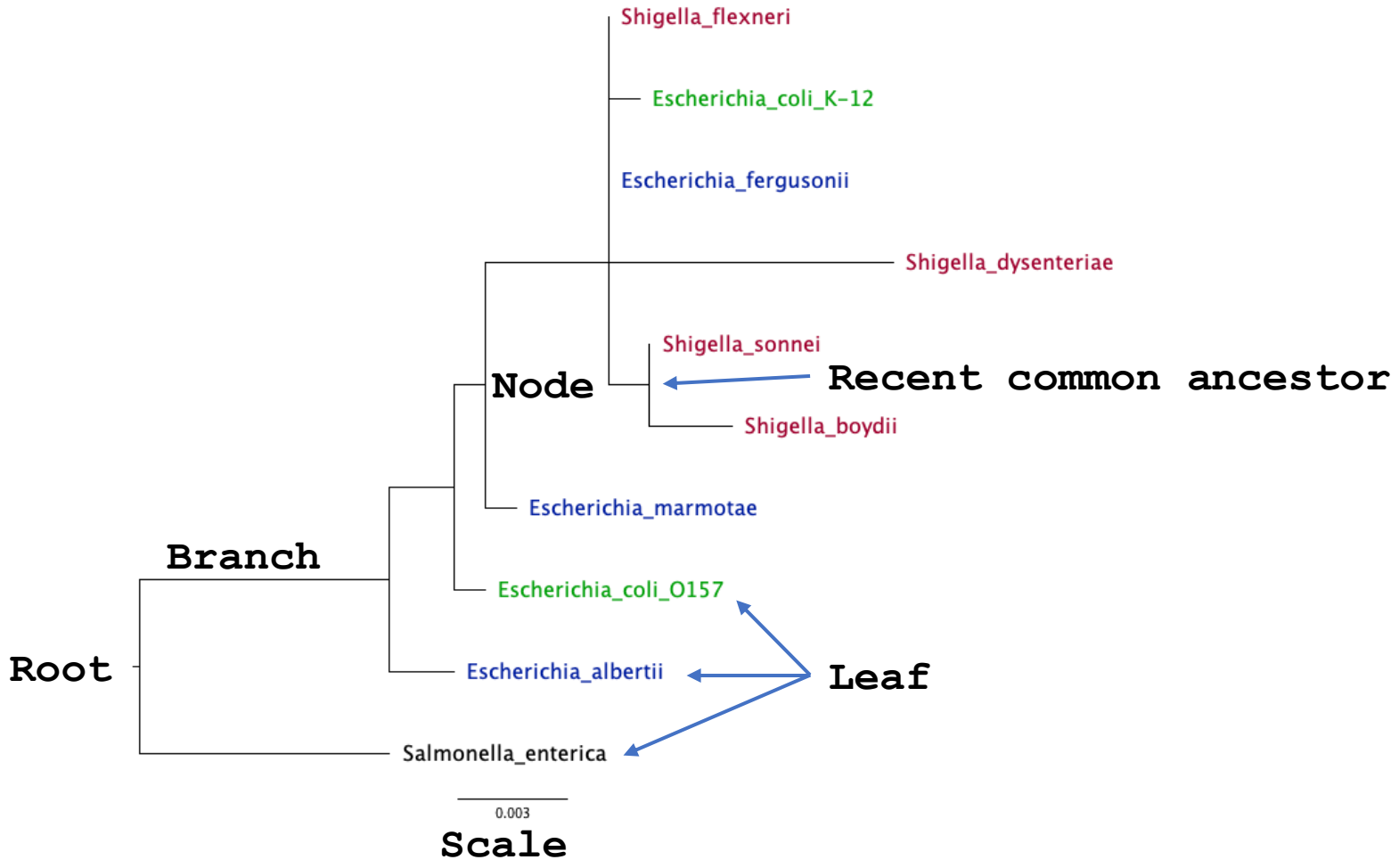
Root

Branch

Leaf



Phylogenetic trees



Practicals

16S rRNA gene sequence analysis and phylogenetics

Sequence data analysis

- Go to https://www.gear-genomics.com/
- Select **`Teal`** and upload your own sequences (`.ab1`) one at a time
  - Inspect how much of bad sequencing data needs to be removed from both ends
  - Can you find heterozygous positions from your data
- Select **`Pearl`** and upload both sequence files
  - Inspect how much the software clipped from both ends
  - How did it resolve possible heterozygous positions
  - Download the sequence: **`Save user sequence (FA)`**

Taxonomic annotation and close relatives

- Go to https://blast.ncbi.nlm.nih.gov/
  - Paste your sequence
  - Select the **`rRNA/ITS databases`** and **`16S ribosomal`**…
  - Hit **`BLAST`**
  - From the results select 10-20 closely related species that you want to add to your tree. You can also select outgroup sequences, if you can find some more distantly related sequences
  - Download the results
  - **Add your own sequence to the file**

Construct phylogenetic tree

- Go to https://www.arb-silva.de/
- Select **`Browser`**
  - Find suitable outgroup sequences. Add them to your cart.
  - When finished, download the cart (options: “fasta without gaps” & “[none]”)
  - Add the sequences to your own sequence file (from blast).
  - Put the furthest outgroup sequence first.

- Select **`ACT`**
  - Paste your sequences and click the **`Compute tree`** tab
  - Use the defaults, or change if you like
  - Name your job and click **`Run tool`**
  - When the job finishes, select it from the list and download the results

Modify the tree

- Go to https://itol.embl.de/
  - Upload your tree and maybe give it a name
  - Re-root the tree from your furthest outgroup, if needed
  - Change the names and modify the tree as you like.
  - When finished, export the tree as PNG (or as SVG if you want to further modify it)